Indian Stock Market Microstructure Analysis

Final Project Report

Internship Capstone Project

Data Science / Machine Learning Track

[Your Name(s)] | [Date]

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1. Executive Summary

This report documents an end-to-end analysis of Indian equity market behavior using historical daily price and volume data for [TICKER] from [START_DATE] to [END_DATE]. The project combines data cleaning, exploratory data analysis, feature engineering, statistical hypothesis testing, and machine learning (clustering and an optional supervised classifier) to describe market behavior and identify distinct market regimes. Key results — to be finalized once the accompanying notebook is executed on live data — include: the statistical significance (or non-significance) of the volume–volatility relationship, the number and character of market regimes identified via clustering, and the degree to which those regimes can be validated using engineered features. This is a descriptive research project; no trading or investment recommendations are made.

2. Introduction

2.1 Motivation

Understanding how markets behave — not just where prices are headed — is foundational to quantitative finance. Market microstructure research examines the mechanics of trading itself: how volume, volatility, and price range interact. This project applies these ideas to Indian equity markets using an accessible, fully reproducible data science workflow suitable for a beginner-to-intermediate practitioner.

2.2 Objectives

Clean and explore historical NIFTY 50 / BANK NIFTY / stock data.

Engineer features that describe daily market behavior.

Statistically test the relationship between volume and volatility.

Discover market regimes via clustering, and validate them via supervised learning.

Communicate findings honestly, with explicit limitations and future directions.

3. Data Description

Column	Description
Date	Trading day (index closed on weekends/exchange holidays)
Open	Price of the first trade of the day
High	Highest price traded during the day
Low	Lowest price traded during the day
Close	Price of the last trade of the day
Adj Close	Close price adjusted for dividends/splits — used for all return calculations
Volume	Total shares/contracts traded during the day

Data source: Yahoo Finance, retrieved via the yfinance Python library. Ticker analyzed: [TICKER]. Period: [START_DATE] – [END_DATE]. Total trading days after cleaning: [N].

4. Methodology

4.1 Data Cleaning

Duplicate rows removed.

Missing values handled via forward-fill (never mean/median-fill or backward-fill, to avoid distorting the time series or introducing lookahead bias).

Rows with logically inconsistent OHLC values (e.g., High < Low) removed.

- Data types verified/coerced to numeric; index sorted chronologically.

4.2 Exploratory Data Analysis

EDA covered univariate distributions (histograms, boxplots), bivariate relationships (price vs. volume, correlation heatmap), time series visualization, and dedicated analysis of returns, volatility, volume, and moving averages. Full details and all charts are in the companion notebook.

4.3 Feature Engineering

- Daily Return and Log Return
- Price Range (absolute and % of price)
- Rolling Mean and Rolling Std (20-day)
- Rolling Volatility (21-day standard deviation of returns)
- Moving Averages (20-day, 50-day)
- Lag Features (1–3-day return lags, volume lag)
- RSI (14-day)

4.4 Hypothesis Testing

Three tests examined whether trading volume relates to the magnitude of price movement: an independent T-test (high vs. low volume days), a one-way ANOVA (volume tertiles), and a Pearson correlation significance test (volume vs. absolute return). All tests used $\alpha = 0.05$.

4.5 Clustering

Features were standardized, then K was chosen via the Elbow Method and Silhouette Score (K = 2–10 evaluated). KMeans and Agglomerative Hierarchical Clustering (Ward linkage) were both fit at the chosen K and compared via a cross-tabulation of cluster assignments.

4.6 Supervised Validation and Evaluation

A Logistic Regression model and a depth-limited Decision Tree were trained to predict cluster labels from same-day features, using a chronological train/test split. Evaluation used accuracy, confusion matrix, and per-class precision/recall/F1.

5. Exploratory Data Analysis — Summary of Findings

- Price trend: [describe overall trend across the period, including any major drawdowns/rallies].

- Daily returns: centered near 0%, with fat tails (extreme days more frequent than a Normal distribution predicts).
- Volatility clustering: visually evident — turbulent and calm periods tended to persist in stretches.
- Volume: right-skewed distribution with a small number of unusually high-activity days.
- Correlation: Open/High/Low/Close/Adj Close near-perfectly correlated (by construction); Volume weakly correlated with price level.

6. Feature Engineering — Summary

Fourteen engineered features were used as the basis for clustering and (optionally) supervised learning: Daily_Return, Log_Return, Price_Range, Price_Range_Pct, Rolling_Mean_20, Rolling_Std_20, Volatility_21d, MA20, MA50, Return_Lag1–3, Volume_Lag1, and RSI_14. All rolling/lag-based NaNs (an expected consequence of needing prior history) were dropped prior to modeling.

7. Hypothesis Testing Results

Test	Statistic	p-value	Conclusion
T-test (high vs. low volume)	[t = ...]	[p = ...]	[Reject / Fail to reject H_0]
ANOVA (volume tertiles)	[F = ...]	[p = ...]	[Reject / Fail to reject H_0]
Pearson correlation (Volume vs. |Return|)	[r = ...]	[p = ...]	[Reject / Fail to reject H_0]

Interpretation: [state in plain language what these three results collectively say about whether higher-volume days see larger price swings in this specific dataset, being careful to distinguish statistical significance from practical magnitude].

8. Clustering Results — Market Regimes

Optimal K (via Elbow Method + Silhouette Score): [K]. Silhouette score at this K: [value]. Agreement between KMeans and Hierarchical Clustering (via crosstab): [strong / moderate / weak].

Cluster	Characteristic Profile	Suggested Label
1	[e.g., high Volatility_21d, wide Price_Range_Pct, RSI near extremes]	[e.g., “Turbulent”]
2	[fill in if K = 3 or more]	[...]

9. Supervised Validation & Model Evaluation

Decision Tree test accuracy: [X]%. Logistic Regression test accuracy: [Y]%. Macro-average F1 (Decision Tree): [value]. Top features by importance: [list top 2–3].

Reminder: this validates that regimes have consistent, learnable same-day feature boundaries — it is not evidence of forecasting ability. A confusion matrix and full classification report are provided in the companion notebook (Milestone 13).

10. Insights and Conclusions

[Write 4–6 sentences synthesizing the whole analysis once real results are available: overall market character during this period, the strength/direction of the volume–volatility relationship, the number and interpretation of regimes found, and how consistently those regimes validated under supervised learning. Keep language descriptive/historical, not predictive or advisory.]

11. Limitations

Daily OHLCV data is used as a simplified proxy for true tick-level market microstructure.

Clusters describe historical patterns only; regime membership is not guaranteed to persist.

Supervised learning here validates same-day structure, not future prediction.

Findings are specific to the analyzed ticker/date range and may not generalize without re-running on other instruments or periods.

12. Future Scope

Intraday/tick-level data for true microstructure analysis (bid-ask spread, order imbalance).

Hidden Markov Models for formal, probabilistic regime-switching (including transition probabilities).

A genuine next-day regime forecasting model with walk-forward validation (TimeSeriesSplit).

Multi-asset / sector comparison across additional NSE-listed instruments.

Incorporating macroeconomic or news-sentiment data to explain regime drivers.

An interactive dashboard (Streamlit/Plotly Dash) for non-technical stakeholders.

13. Appendix

13.1 Tech Stack

Python, pandas, numpy, matplotlib, seaborn, scikit-learn, scipy, yfinance, plotly, Jupyter Notebook

13.2 Repository Structure

See README.md for the full folder structure and setup instructions.

13.3 Reproducibility

All results in this report can be reproduced by running Indian_Stock_Market_Microstructure_Analysis.ipynb top-to-bottom with an active internet connection and the dependencies listed in requirements.txt.